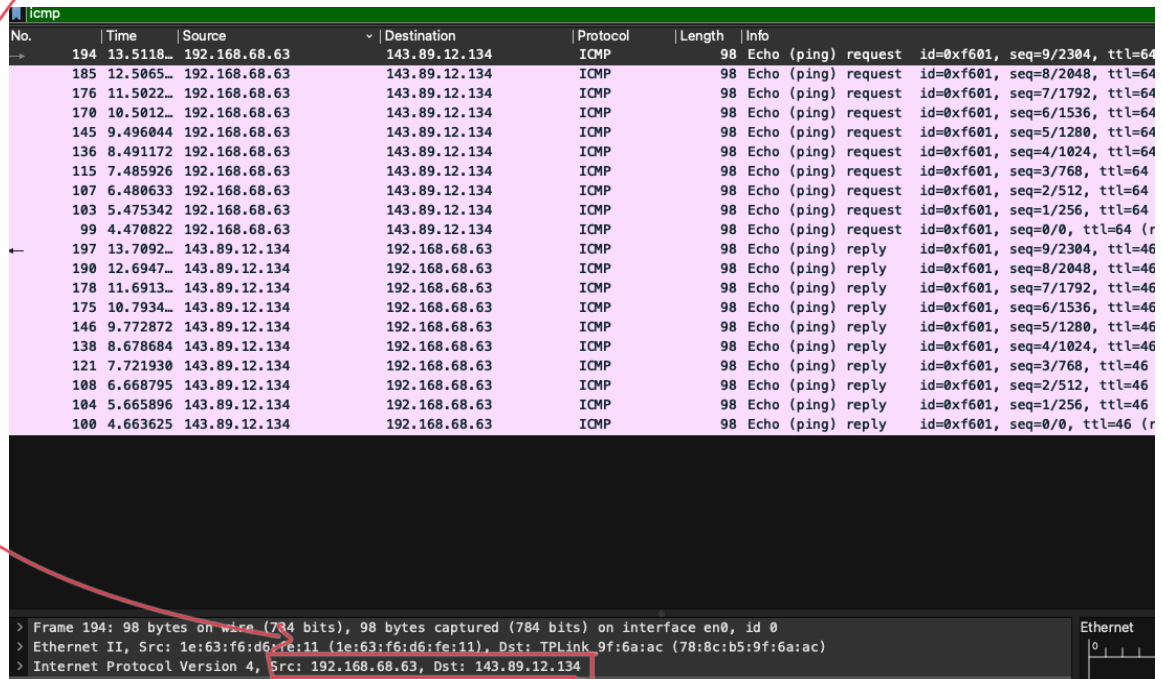


Lab 8 ICMP

1. What is the IP address of your host? What is the IP address of the destination Host?

Host IP: 192.168.68.63, Dest IP: 143.89.12.134



No.	Time	Source	Destination	Protocol	Length	Info
194	13.5118...	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=9/2304, ttl=64
185	12.5065...	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=8/2048, ttl=64
176	11.5022...	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=7/1792, ttl=64
170	10.5012...	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=6/1536, ttl=64
145	9.496044	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=5/1280, ttl=64
136	8.491172	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=4/1024, ttl=64
115	7.485926	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=3/768, ttl=64
107	6.480633	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=2/512, ttl=64
103	5.475342	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=1/256, ttl=64
99	4.470822	192.168.68.63	143.89.12.134	ICMP	98	Echo (ping) request id=0xf601, seq=0/0, ttl=64 (r
197	13.7092...	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=9/2304, ttl=46
190	12.6947...	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=8/2048, ttl=46
178	11.6913...	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=7/1792, ttl=46
175	10.7934...	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=6/1536, ttl=46
146	9.772872	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=5/1280, ttl=46
138	8.678684	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=4/1024, ttl=46
121	7.721930	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=3/768, ttl=46
108	6.668795	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=2/512, ttl=46
104	5.665896	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=1/256, ttl=46
100	4.663625	143.89.12.134	192.168.68.63	ICMP	98	Echo (ping) reply id=0xf601, seq=0/0, ttl=46 (r

> Frame 194: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
 > Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11), Dst: TPLink 9f:6a:ac (78:8c:b5:9f:6a:ac)
 > Internet Protocol Version 4, Src: 192.168.68.63, Dst: 143.89.12.134

2. Why is it that an ICMP packet does not have source and destination Port Numbers?

Firstly ICMP is not a transport protocol(UDP, TCP), ICMP is used for network diagnostics and error reporting, not for delivering data between applications. Secondly, ICMP is used for network communication, not application communication, ports are only needed when distinguishing between different applications, which is not ICMP's purpose, for example, Echo request, we did are meant for devices to communicate network status. Also, ICMP messages are directly encapsulated in IP, ICMP is encapsulated directly within IP, skipping the transport layer, so there's no concept of port numbers. Finally, ICMP Uses Different Identifiers Instead of Ports.

3. Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

ICMP type: 8 (Echo (ping) request), and coded number: 0;

Other fields are: checksum, identifier (BE and LE), sequence number (BE and LE), timestamp;

The checksum, identifier, and sequence number each take 2 bytes.

The screenshot shows a Wireshark packet capture of ICMP ping traffic. The top pane displays a list of packets, with frame 194 selected. The bottom pane shows the details of frame 194, which is an ICMP Echo (ping) request. The details are as follows:

```

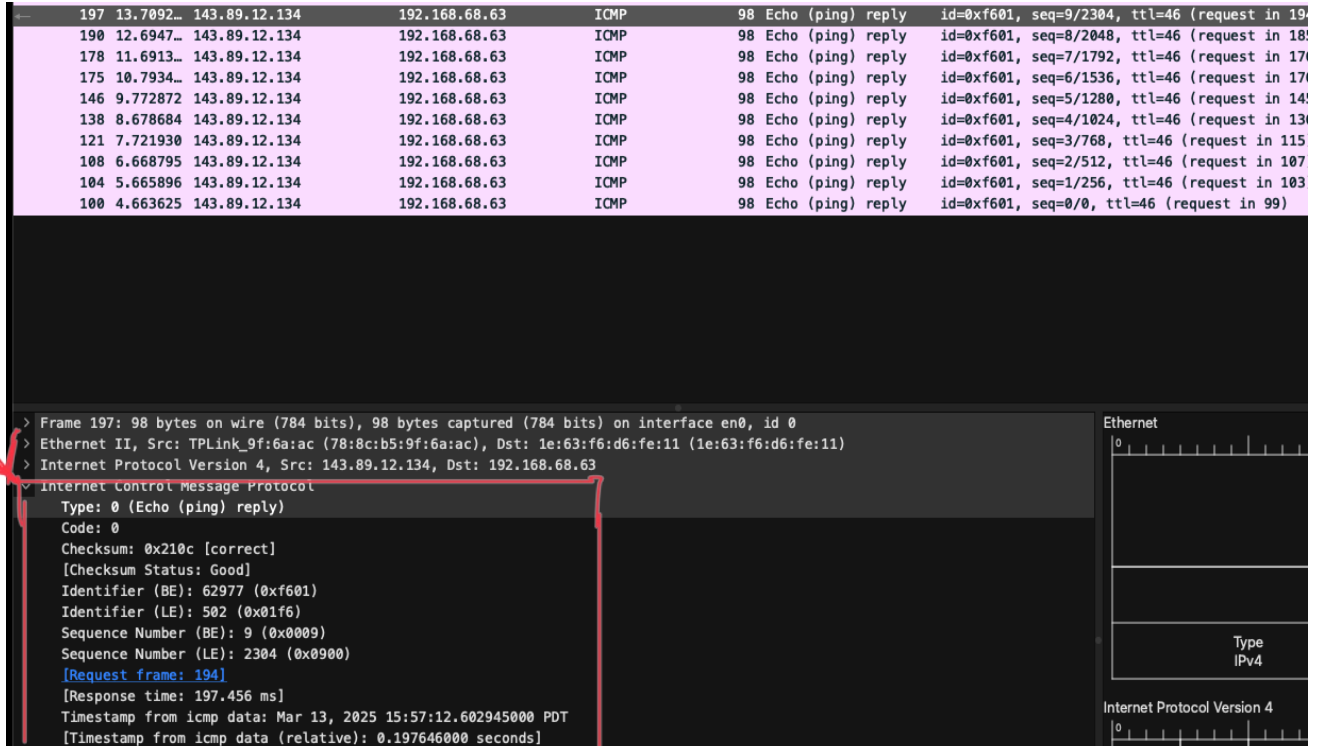
> Frame 194: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
Internet Protocol Version 4, Src: 192.168.68.63, Dst: 143.89.12.134
Internet Control Message Protocol
  Type: 8 (Echo (ping) request)
  Code: 0
  Checksum: 0x190c [correct]
  [Checksum Status: Good]
  Identifier (BE): 62977 (0xf601)
  Identifier (LE): 502 (0x01f6)
  Sequence Number (BE): 9 (0x0009)
  Sequence Number (LE): 2304 (0x0900)
  [Response frame: 197]
  Timestamp from icmp data: Mar 13, 2025 15:57:12.602945000 PDT
  [Timestamp from icmp data (relative): 0.000190000 seconds]
Data (48 bytes)
  Data: 08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637
  
```

4. Examine the corresponding ping reply packet. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

The reply packet of ICMP type: Type: 0 (Echo (ping) reply) and Code: 0

Other fields are: checksum, identifier (BE and LE), sequence number (BE and LE), responde time, timestamp;

The checksum, identifier, and sequence number each take 2 bytes.



197 13.7092... 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=9/2304, ttl=46 (request in 19)

190 12.6947... 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=8/2048, ttl=46 (request in 18)

178 11.6913... 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=7/1792, ttl=46 (request in 17)

175 10.7934... 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=6/1536, ttl=46 (request in 17)

146 9.772872 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=5/1280, ttl=46 (request in 14)

138 8.678684 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=4/1024, ttl=46 (request in 13)

121 7.721930 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=3/768, ttl=46 (request in 11)

108 6.668795 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=2/512, ttl=46 (request in 10)

104 5.665896 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=1/256, ttl=46 (request in 10)

100 4.663625 143.89.12.134 192.168.68.63 ICMP 98 Echo (ping) reply id=0xf601, seq=0/0, ttl=46 (request in 9)

> Frame 197: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0

> Ethernet II, Src: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11)

> Internet Protocol Version 4, Src: 143.89.12.134, Dst: 192.168.68.63

> Internet Control Message Protocol

Type: 0 (Echo (ping) reply)

Code: 0

Checksum: 0x210c [correct]

[Checksum Status: Good]

Identifier (BE): 62977 (0xf601)

Identifier (LE): 502 (0x01f6)

Sequence Number (BE): 9 (0x0009)

Sequence Number (LE): 2304 (0x0900)

[Request frame: 194]

[Response time: 197.456 ms]

Timestamp from icmp data: Mar 13, 2025 15:57:12.602945000 PDT

[Timestamp from icmp data (relative): 0.197646000 seconds]

Ethernet

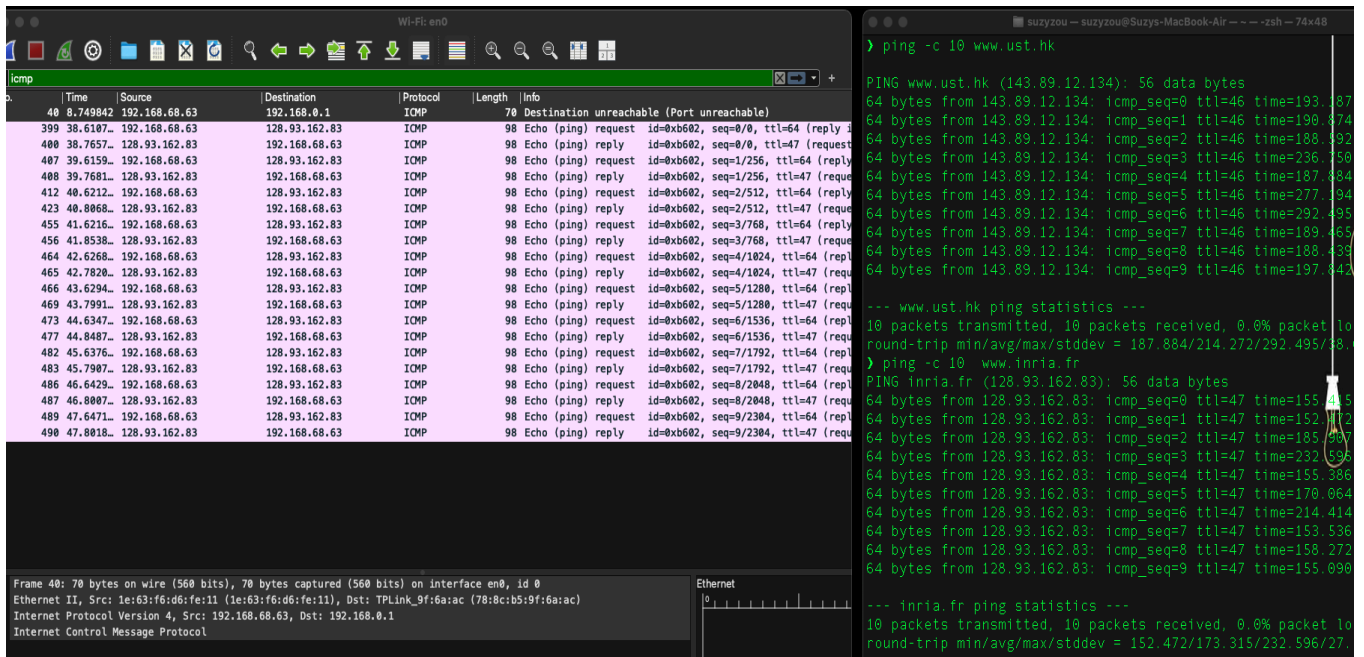
Type IPv4

Internet Protocol Version 4

ICMP and Traceroute

5. What is the IP address of your host? What is the IP address of the target destination host?

IP host: 192.168.68.63, destination IP: 128.93.162.83



Wi-Fi: en0

No.	Time	Source	Destination	Protocol	Length	Info
399	0.749842	192.168.68.63	128.93.162.83	ICMP	70	Destination unreachable (Port unreachable)
400	38.6107...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=0/0, ttl=64 (reply 1)
401	38.7657...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=0/0, ttl=47 (request 1)
402	39.6159...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=1/256, ttl=64 (reply 1)
403	39.7681...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=1/256, ttl=47 (request 1)
412	40.6212...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=2/512, ttl=64 (reply 1)
423	40.8068...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=2/512, ttl=47 (request 1)
455	41.6216...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=3/768, ttl=64 (reply 1)
456	41.8538...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=3/768, ttl=47 (request 1)
464	42.6268...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=4/1024, ttl=64 (reply 1)
465	42.7828...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=4/1024, ttl=47 (request 1)
466	43.6294...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=5/1280, ttl=64 (reply 1)
469	43.7991...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=5/1280, ttl=47 (request 1)
473	44.6347...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=6/1536, ttl=64 (reply 1)
477	44.8487...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=6/1536, ttl=47 (request 1)
482	45.6376...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=7/1792, ttl=64 (reply 1)
483	45.7987...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=7/1792, ttl=47 (request 1)
486	46.6429...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=8/2048, ttl=64 (reply 1)
487	46.8087...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=8/2048, ttl=47 (request 1)
489	47.6471...	192.168.68.63	128.93.162.83	ICMP	98	Echo (ping) request id=0xb602, seq=9/2304, ttl=64 (reply 1)
490	47.8018...	128.93.162.83	192.168.68.63	ICMP	98	Echo (ping) reply id=0xb602, seq=9/2304, ttl=47 (request 1)

Frame 40: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0

Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)

Internet Protocol Version 4, Src: 192.168.68.63, Dst: 128.93.162.83

Internet Control Message Protocol

Wireshark

suzyzou@suzyzou-MacBook-Air ~ - ssh - 74x48

```
> ping -c 10 www.ust.hk
PING www.ust.hk (143.89.12.134): 56 data bytes
64 bytes from 143.89.12.134: icmp_seq=0 ttl=46 time=193.887 ms
64 bytes from 143.89.12.134: icmp_seq=1 ttl=46 time=190.874 ms
64 bytes from 143.89.12.134: icmp_seq=2 ttl=46 time=188.192 ms
64 bytes from 143.89.12.134: icmp_seq=3 ttl=46 time=236.758 ms
64 bytes from 143.89.12.134: icmp_seq=4 ttl=46 time=187.884 ms
64 bytes from 143.89.12.134: icmp_seq=5 ttl=46 time=277.194 ms
64 bytes from 143.89.12.134: icmp_seq=6 ttl=46 time=292.495 ms
64 bytes from 143.89.12.134: icmp_seq=7 ttl=46 time=189.465 ms
64 bytes from 143.89.12.134: icmp_seq=8 ttl=46 time=188.438 ms
64 bytes from 143.89.12.134: icmp_seq=9 ttl=46 time=197.443 ms

--- www.ust.hk ping statistics ---
10 packets transmitted, 10 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 187.884/214.272/292.495/38.111 ms

> ping -c 10 www.inria.fr
PING inria.fr (128.93.162.83): 56 data bytes
64 bytes from 128.93.162.83: icmp_seq=0 ttl=47 time=155.413 ms
64 bytes from 128.93.162.83: icmp_seq=1 ttl=47 time=152.112 ms
64 bytes from 128.93.162.83: icmp_seq=2 ttl=47 time=185.997 ms
64 bytes from 128.93.162.83: icmp_seq=3 ttl=47 time=232.596 ms
64 bytes from 128.93.162.83: icmp_seq=4 ttl=47 time=155.386 ms
64 bytes from 128.93.162.83: icmp_seq=5 ttl=47 time=170.064 ms
64 bytes from 128.93.162.83: icmp_seq=6 ttl=47 time=214.414 ms
64 bytes from 128.93.162.83: icmp_seq=7 ttl=47 time=153.536 ms
64 bytes from 128.93.162.83: icmp_seq=8 ttl=47 time=158.272 ms
64 bytes from 128.93.162.83: icmp_seq=9 ttl=47 time=155.090 ms

--- inria.fr ping statistics ---
10 packets transmitted, 10 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 152.472/173.315/232.596/27.111 ms
```

6. If ICMP sent UDP packets instead (as in Unix/Linux), would the IP protocol number still be 01 for the probe packets? If not, what would it be?

If ICMP were to send UDP packets instead, the IP protocol number would not be 01. Instead, it would be 17 (0x11 in hexadecimal), which is the designated protocol number for UDP in the IP header.

7. Examine the ICMP echo packet in your screenshot. Is this different from the ICMP ping query packets in the first half of this lab? If yes, how so?

From my observation, the first half and the second half are similar, because in the ICMP fields are type, code, checksum, identifier(BE, LE), sequence number(BE, LE) respond time, and timestamp from the icmp data.

The screenshot shows a Wireshark packet capture of ICMP Echo (ping) traffic. The top pane displays a list of packets, including several Echo (ping) requests and replies. A red arrow points to the details of the first packet (Frame 399), which is an Echo (ping) request. The details pane shows the following fields:

- Frame 399: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
- Ethernet II, Src: 1e:63:f6:d6:fe:11 (1e:63:f6:d6:fe:11), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
- Internet Protocol Version 4, Src: 192.168.68.63, Dst: 128.93.162.83
- Internet Control Message Protocol
 - Type: 8 (Echo (ping) request)
 - Code: 0
 - Checksum: 0x65a6 [correct] [Checksum Status: Good]
 - Identifier (BE): 46594 (0xb602)
 - Identifier (LE): 694 (0x02b6)
 - Sequence Number (BE): 0 (0x0000)
 - Sequence Number (LE): 0 (0x0000)
 - [Response frame: 400]
 - Timestamp from icmp data: Mar 13, 2025 16:22:48.467121000 PDT
 - [Timestamp from icmp data (relative): 0.000150000 seconds]
- Data (48 bytes)
 - Data: 00090a0b0c0d0e0f011112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637
 - [Length: 48]

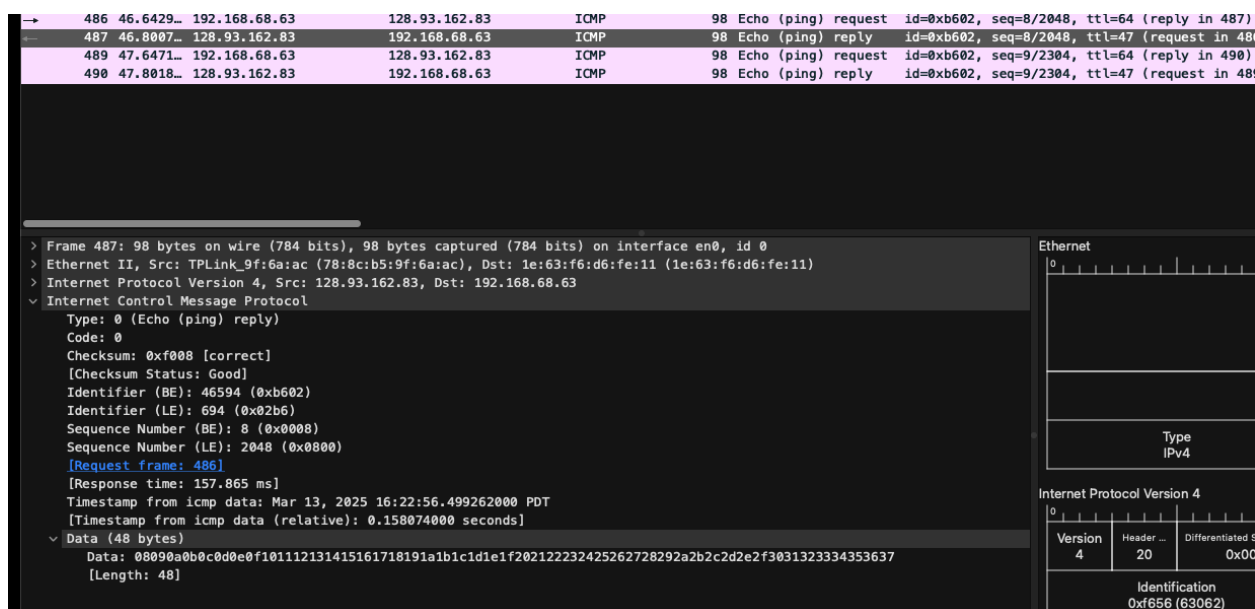
The right pane shows the packet structure for Ethernet, Internet Protocol Version 4, and Internet Control Message Protocol.

8. Examine the ICMP error packet in your screenshot. It has more fields than the ICMP echo packet. What is included in those fields?

The ICMP error packet, specifically a "Time-to-Live Exceeded" message (Type 11, Code 0), includes an unused field reserved for future use, typically set to zero. It also contains a nested IP header along with the initial portion of the triggering packet. This snapshot of the original packet—an ICMP Echo Request in this case—enables the sender to identify which packet encountered an issue along its route.

9. Examine the last three ICMP packets received by the source host. How are these packets different from the ICMP error packets? Why are they different?

The last three ICMP packets in the traceroute results are responses from both intermediate routers and the destination host. Unlike ICMP error packets, these responses may include other ICMP message types, such as "Echo Reply" or additional "Time-to-Live Exceeded" messages, depending on the hop and the TTL status. Their purpose extends beyond indicating TTL expiration, as they provide various network-related information or error notifications.



10. Within the tracert measurements, is there a link whose delay is significantly longer than others? Refer to the screenshot in Figure 4, is there a link whose delay is significantly longer than others? On the basis of the router names, can you guess the location of the two routers on the end of this link?

A noticeable delay occurs between hops 6 and 7 in the traceroute results, where the round-trip time increases from approximately 83 ms to 194 ms. This suggests a geographical transition and possibly a bandwidth-related bottleneck. The router at hop 6 (196.250.236.246) does not reveal much geographic information but is likely located within the same region or country. By hop 8 (100ge0-79.core1.lis1.he.net), the packet has clearly reached Lisbon, Portugal, as indicated by the router's hostname.